Project ID: 25-26J-106

ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Main Supervisor
Mr. Samadhi Rathnayake
Lecturer
Faculty of Computing SLIIT

Co-Supervisor
Mr. Amila Nuwan Alexander
Assistant Lecturer
Faculty of Computing SLIIT

COLLABORATIVE TEXTURE RECOGNITION

Specialization: Data Science

COLLABORATIVE TEXTURE RECOGNITION

Problem

No existing system reliably sorts glass, plastic, rubber, and metal in real time using vision, sensors, and human feedback.

User Requirements Addressed by the Solution

- Reliable detection of multiple materials (glass, plastic, rubber, metal)
- Live-stream capability for real-time recognition.
- Adaptability
- Fast
- Performance

Solution

Camera (Visual Data) → DINOv2 (ViT)

Ultrasonic sensor (Distance Signals) → Sensor Feature Extraction

→ Multimodal Classification Model →

Material Classification

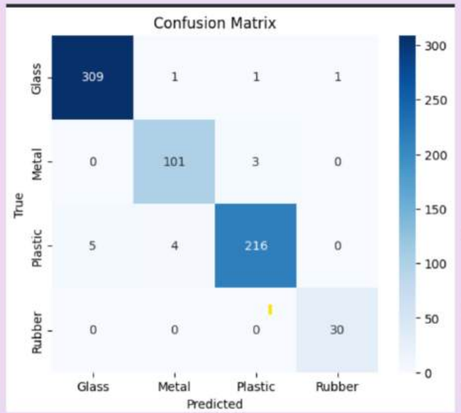
Design

- Multimodal Sensing: Combines vision (DINOv2) and ultrasonic sensor data for robust material recognition.
- Adaptive Model: Handles materials with no public datasets, such as rubber.
- Real-Time Performance: Enables live-stream material classification for practical applications.
- Future-Ready: Supports human-in-the-loop feedback and reinforcement learning.
- Flexible & Scalable: Easily extended to new sensors or material types.

ADAPTIVE PATTERN RECOGNITION VIA HUMAN BEHAVIOR

Specialization: Data Science

Proof of Concept



DINOv2

✅ Test Accuracy: 97.76%

CNN Models

- **Sequential** → 44.19%
- **EfficientNet** → 87.72%
- **MobileNet** → 88.72%

```
Classification Report:

              precision    recall  f1-score   support

       Glass       0.98      0.99      0.99       312
       Metal       0.95      0.97      0.96       104
     Plastic       0.98      0.96      0.97       225
      Rubber       0.97      1.00      0.98        30

    accuracy                           0.98       671
   macro avg       0.97      0.98      0.98       671
weighted avg       0.98      0.98      0.98       671
```

Why DINOv2

Because it learns strong visual representations without labels, making it highly effective for texture and material recognition with limited data.

Technologies

Flask — web development, one drop at a time
OpenCV
Meta DINOv2
PyTorch

Applications

- Automated material sorting
- Human–robot collaboration
- Recycling and waste processing
- Manufacturing assembly lines
- Robotic handling and inspection systems
- Electronics Manufacturing & E-Waste
- Apparel & Textile Industry

Risk Mitigation

- Dataset gaps → Manual data collection
- Live accuracy issues → DINOv2 model
- Sensor unreliability → Ultrasonic sensor
- Generalization risk → Human feedback

Feedback

- Understandable predictions
- Improved user confidence
- Better real-time interaction

Adaptive Pattern Recognition via Human Behavior

Problem

Object pattern sorting is inefficient and struggles with manual handling and **no human-robot collaboration**.

Requirements

Enable real-time object sorting and voice-command detection for robotic execution.

Performance
Usability
Adaptability

Solution

Real Time Object Detection

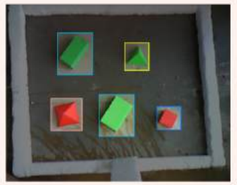
Human Pattern Detection

Human voice Detection

Robot Sorting Strategy

Sort

Design



Annotation objects & hand
Roboflow

YoloV11
Training

Video data frames + Logical Pattern Detection

Sort

Command Dataset Creation

Logistic Regression

Adaptive Pattern Recognition via Human Behavior

Proof of Concept

Metric	Value
Precision	87%
Recall	81.7%
mAP@0.50	90%
mAP@0.50–0.95	70%

YoloV11

Logistic Regression - 96%
Naive Bayes - 88%
decision tree - 64%

Why Logistic Regression - NLP

interpretable, and reliable classification for short, structured voice commands, while requiring minimal computational resources.

Technologies

YOLOv11

Logistic Regression

OpenCV

Applications

- Logistics & Warehousing
- Manufacturing & Assembly Lines
- Recycling & Waste Management
- Agricultural Processing
- Pharmaceutical Packaging
- Electronics & E-Waste Handling
- Textile & Apparel Industry
- Food Processing & Packaging

Risk Mitigation

- Mannual Data Collection and Data Set Creation
- Live Data Streaming with Controlled Data Capture

Feedback

- Clear and logical pattern recognition flow
- Understandability

HUMAN-LED SELECTION THROUGH GAZE-GUIDED ROBOTIC ASSISTANCE

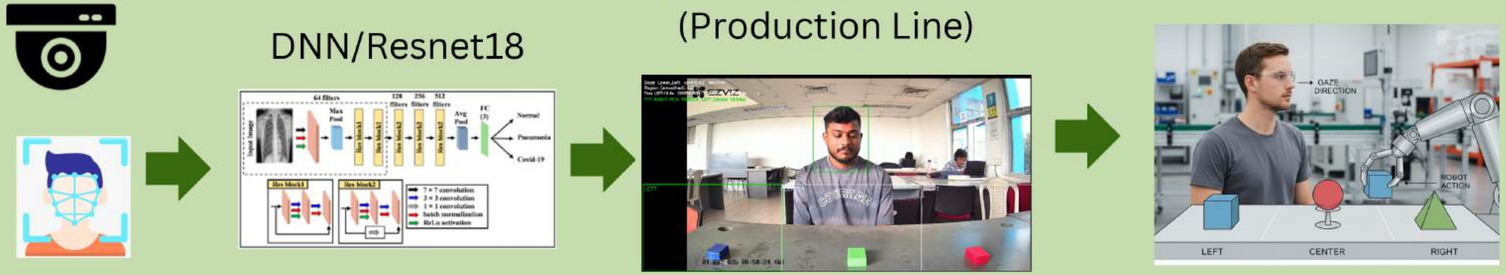
Problem

Robots lack a natural, hands-free way to understand human intent

Solution

DNN/Resnet18

Region Mapping
With Gaze Direction
(Production Line)



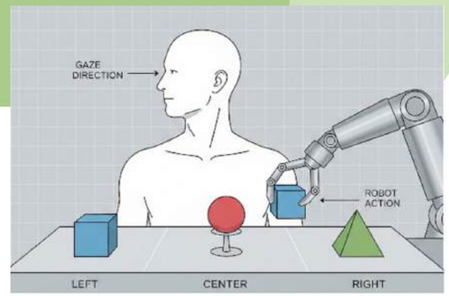
Our solution uses a standard RGB camera to capture human gaze and interpret intent without any wearable or intrusive hardware.”

Requirements

- Hands-free, natural interaction
- No wearable or intrusive hardware
- Safe activation through stable gaze
- Works in structured production lines
- **Security, Reliability, Usability**

User Feedback

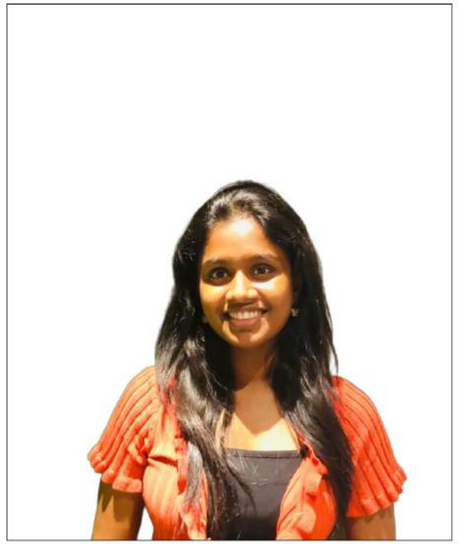
- Intuitive and easy to use
- Reduced accidental activation
- No wearable or intrusive hardware



Design

- Appearance-based gaze classification
- Region-level gaze abstraction suitable for human–robot interaction
- Face Detection → Gaze Classification → Region Mapping → Robot Trigger
- Designed for generalization and real-time deployment

Your paragraph text

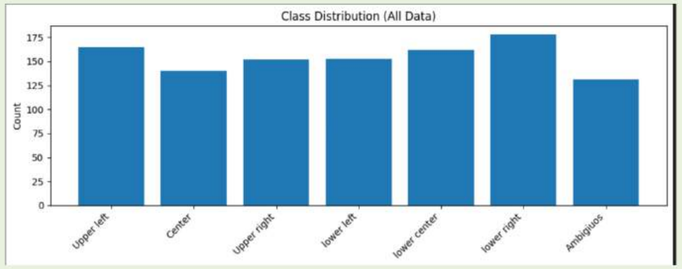


HUMAN-LED SELECTION THROUGH GAZE-GUIDED ROBOTIC ASSISTANCE

Specialization: Data Science

HUMAN-LED SELECTION THROUGH GAZE-GUIDED ROBOTIC ASSISTANCE

Proof of Concept



- ✅ Test loss: 0.7601
- ✅ Test acc : 0.8650
- ✅ Macro-F1 : 0.8643

	precision	recall	f1-score	support
Upper left	0.8750	0.8400	0.8571	25
Center	1.0000	0.9048	0.9500	21
Upper right	0.9565	0.9565	0.9565	23
lower left	0.8000	0.8696	0.8333	23
lower center	0.7778	0.8400	0.8077	25
lower right	0.8889	0.8889	0.8889	27
Ambigiuos	0.7778	0.7368	0.7568	19
accuracy			0.8650	163
macro avg	0.8680	0.8624	0.8643	163
weighted avg	0.8681	0.8650	0.8657	163

Technologies

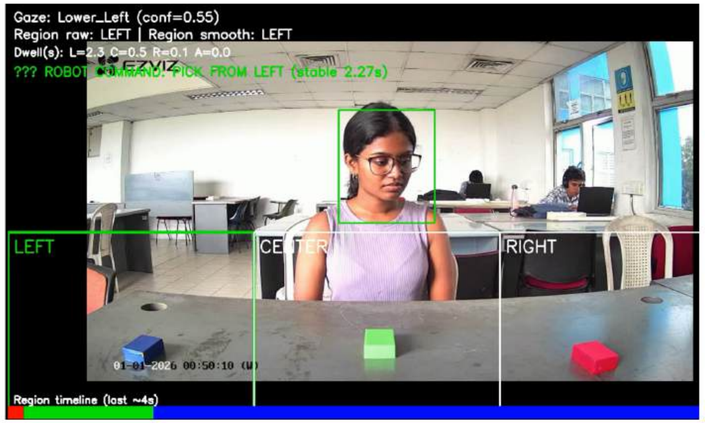
Flask, OpenCV, Meta, PyTorch

Why ResNet18 ?

provides the best balance of accuracy, generalization, and efficiency

Real World Impact

Bridges the gap between gaze estimation research and real-world robot interaction.



Risk Mitigation

- Gaze misclassification → Temporal smoothing and dwell-time threshold
- Accidental robot activation → Stable gaze requirement before trigger
- Poor lighting or occlusion → Ignore uncertain or no-face detections

Applications

- Industrial production lines
- Assembly and packaging systems
- Human–robot collaboration (HRC)
- Smart manufacturing
- Assistive robotics

INTEGRATING EXPLAINABLE AI AND ADAPTIVE RECOMMENDATIONS FOR HUMAN-CENTERED ROBOTIC SORTING

Specialization: Data Science

Problem

Manual object-based decision making is **slow**, inconsistent, and **not explainable** in real-time systems.

Requirements

Real-time recommendations
High Accuracy and Fast
Transparent, explainable decisions

Performance
Usability
Adaptability
Scalability

Solution

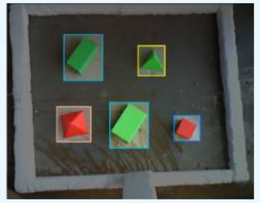
Real Time Object Detection

Generate Recomendations

Explain

Sort

Design & Key Pillars



Annotation
Roboflow
600 Images

YOLOv8

Training

Detection
ESP32 Camera

Feature Extraction

Random Forest

XGBoost

Logistic Regression

2500 Dataset
Recommendations

SHAP
Global, local, and Operational

Model Explanation

6DOF
Robo Arm

Flask

Proof of Concept

YOLOv8

Metric	Value
Precision	99%
Recall	99.8%
mAP@0.50	99.2%
mAP@0.50–0.95	70%

Accuracy

Random Forest	0.87
XGBoost	0.88
Logistic Regression	0.88

Why Random Forest

For a real-time robotic system, consistency and reliability. And good for SHAP

Technologies

Flask web development, one drop at a time

OpenCV

Shap

YOLOv8

Applications

- Logistics & Warehousing
- Manufacturing Sorting Station
- Recycling & Waste Segregation Plants
- Post-harvest processing centers
- Pharmaceutical & Medical Packaging
- Electronics Manufacturing & E-Waste
- Apparel & Textile Industry
- Food Processing & Packaging

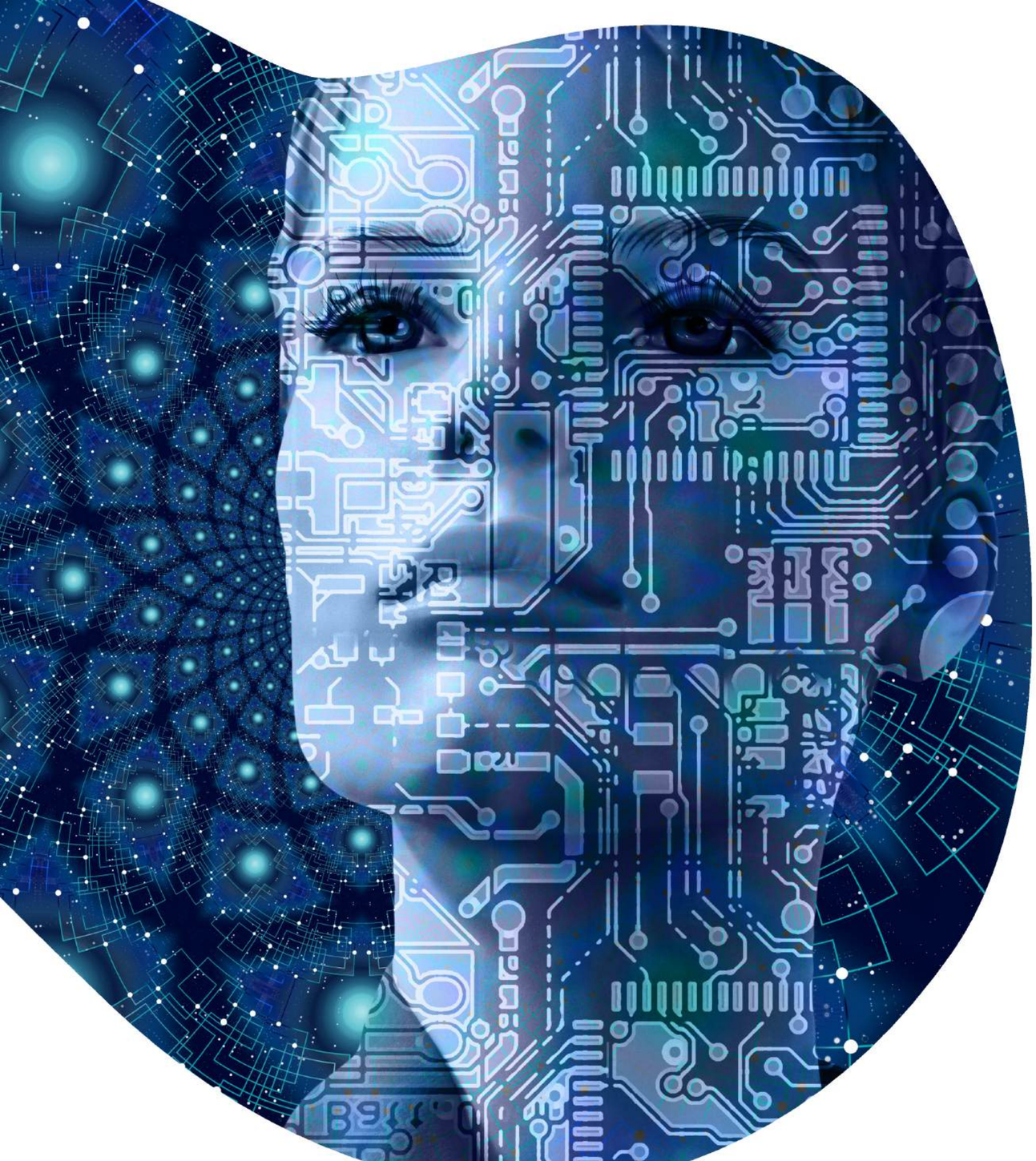
Risk Mitigation

- Evaluation Metrics
- Data Set Creation over real incidents
- Smart Recommendation providing

Feedback

- Understandable
- Explanations improve confidence
- Faster decision-making.

Demonstration



THANK YOU